

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

HYDROCARBONS

LECTURE-06 CHEMISTRY

VISHAL SIR

JKBose Wallah

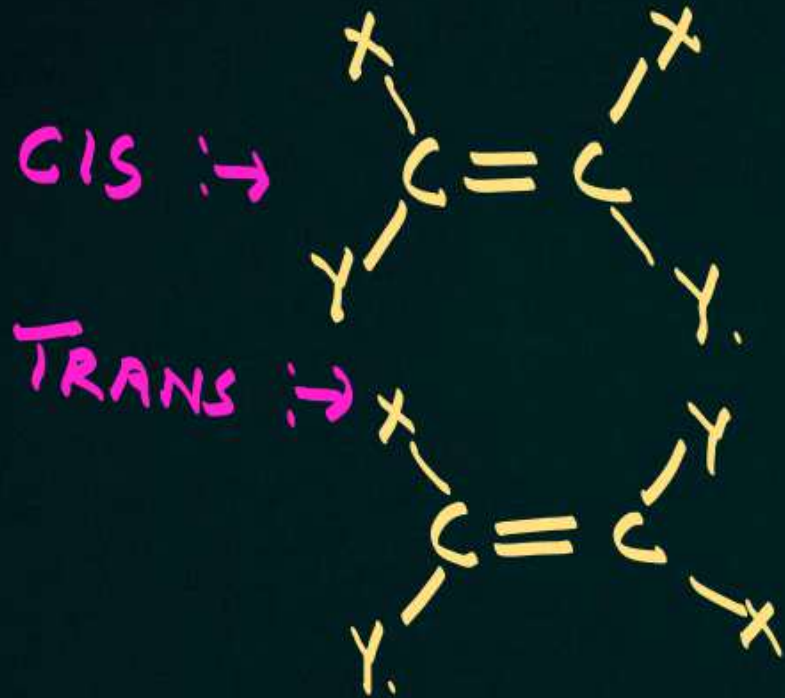


Topics To Covered

- 1 Preparations of Alkenes**
- 2 Physical Properties of Alkenes**
- 3 Chemical Properties of Alkenes**

REVISION

GEOMETRICAL ISOMERISM



ALKENES KO BANANA KA



A



From Alkyne



Trans.

Dehydration of Alcohol.

A



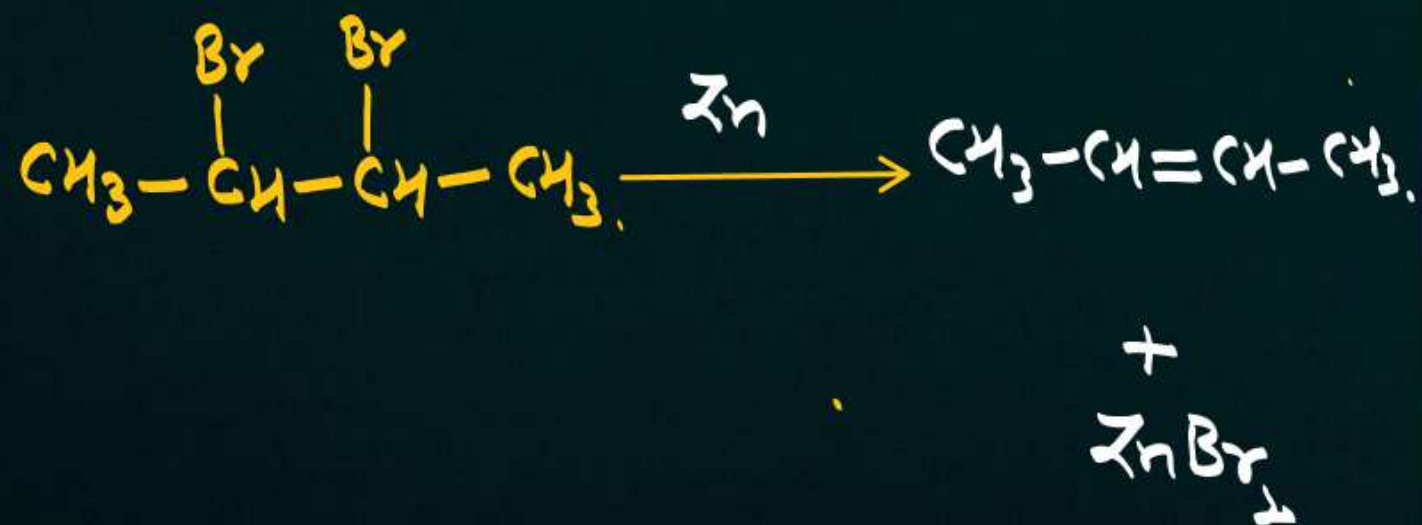
$\rightarrow$ β -Elimination
 $\rightarrow$ dehydro halogenation
 $\rightarrow$ Saytzeff Rule.

Preparations of Alkenes

3. From vicinal dihalides:

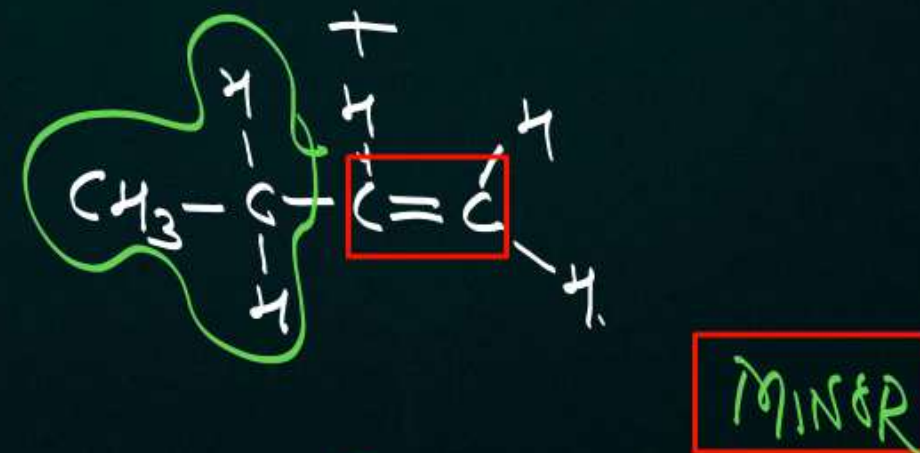
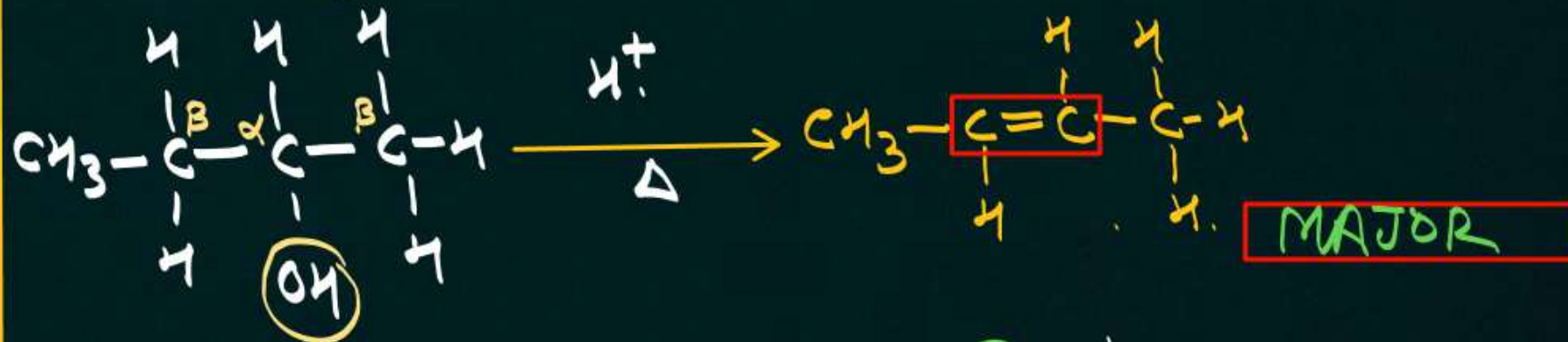
Dihalides having halogens present on adjacent carbon are called vicinal dihalides.

Vicinal dihalides are converted into alkenes when treated with Zn.

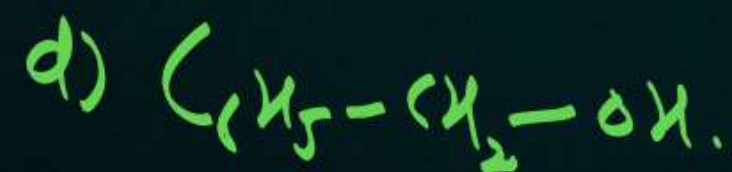
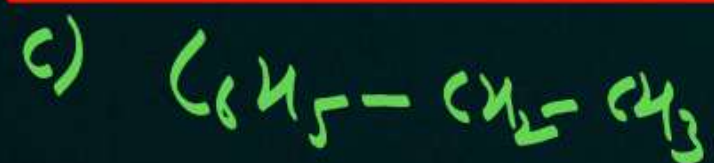
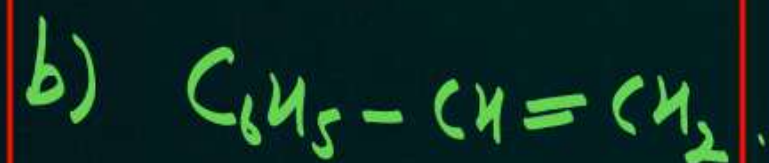
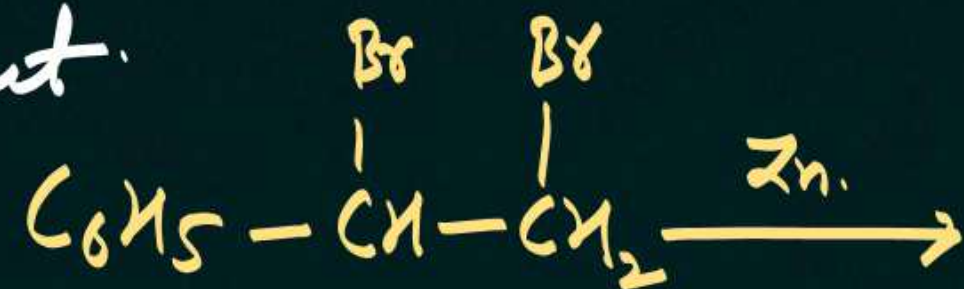


4. From acidic dehydration of alcohol:

Alcohols having β -hydrogen on treatment of acid, undergo β -elimination to form alkene according to Saytzeff rule.



Q. Identify the product.



Physical Properties of Alkenes

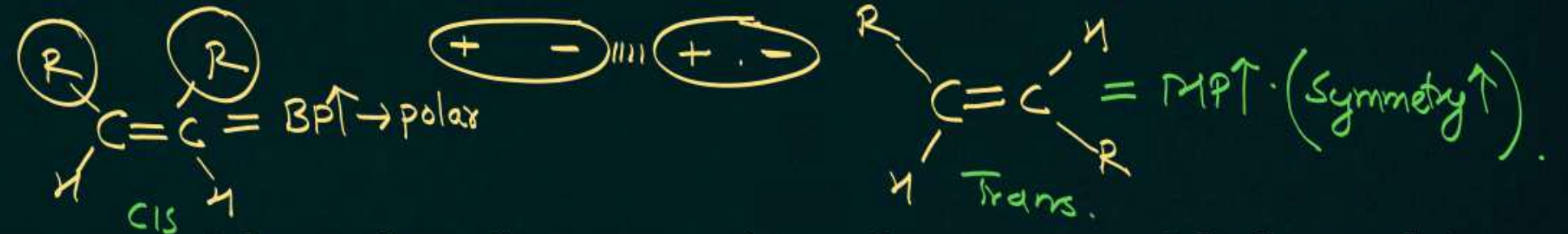
1-3 → Gases
4-17 → Liquids
18 → Solids.

(a) State, Colour & Odour:

The first three alkenes are gases, the next fourteen are liquids and higher members are solids. They are colourless and odourless, except ethene which has a faint sweet smell.

(b) Solubility:

Alkenes are insoluble in water but dissolve in non-polar solvents like benzene, ether and CCl_4 .



(c) Boiling & Melting Points:

Boiling and melting points increase with molecular mass but decrease with branching. Cis-isomers have higher boiling points, while trans-isomers have higher melting points.

(d) Polarity & Density:

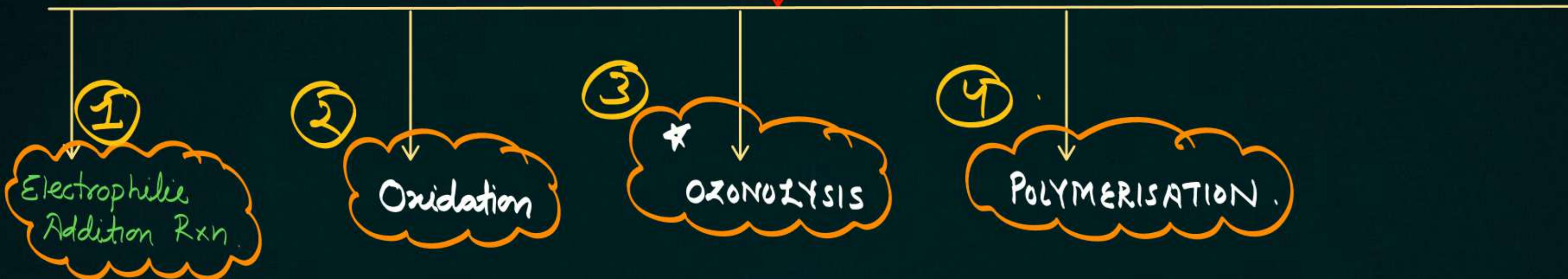
Stability: $\text{Trans alkene} > \text{Cis alkene}$

Trans-isomers have almost zero dipole moment and weaker intermolecular forces. Alkenes are lighter than water and their density increases with molecular mass.

Chemical Properties of Alkenes



Σ 00 P



- i) Addition of H_2
- ii) " " X_2
- iii) " " HX
- iv) " " H_2SO_4
- v) " " H_2O

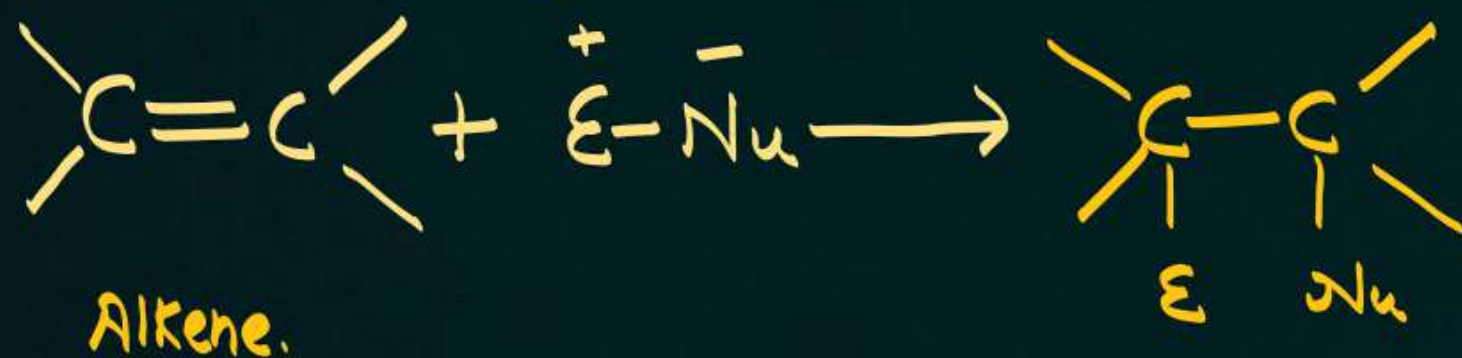
Chemical Properties of Alkenes



Mechanism (Ionic mechanism).

1) Electrophilic Addition Reaction $\rightarrow$

General electrophilic addition Rxn may be represented by.



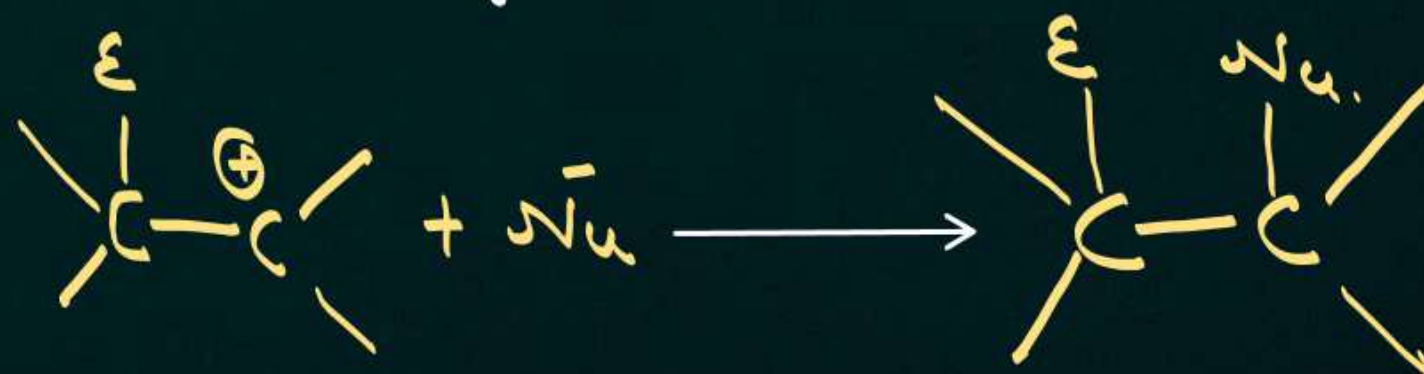
a). Generation of electrophile.



b) Formation of carbocation



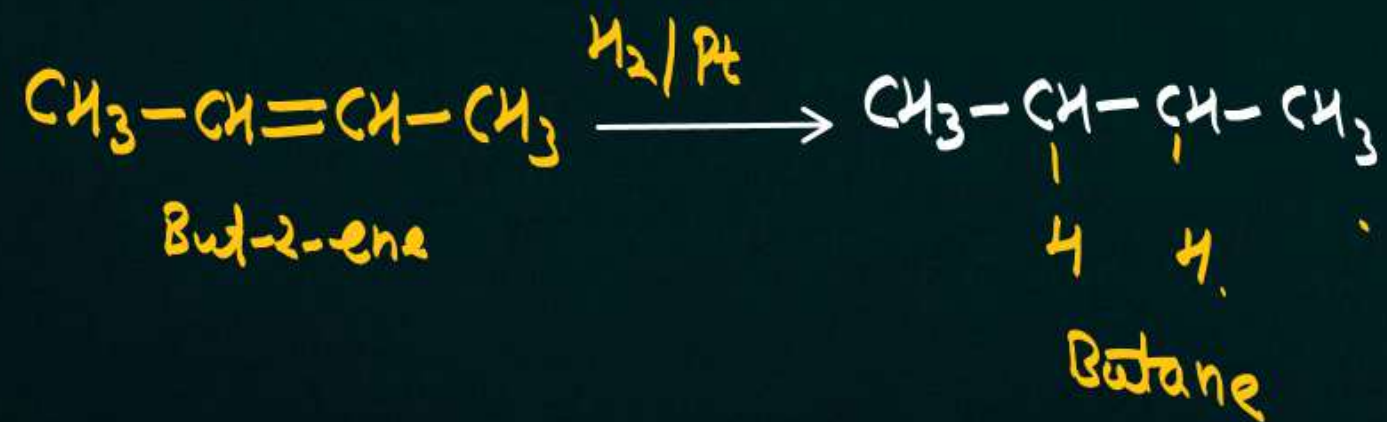
c) Formation of Product



Chemical Properties of Alkenes

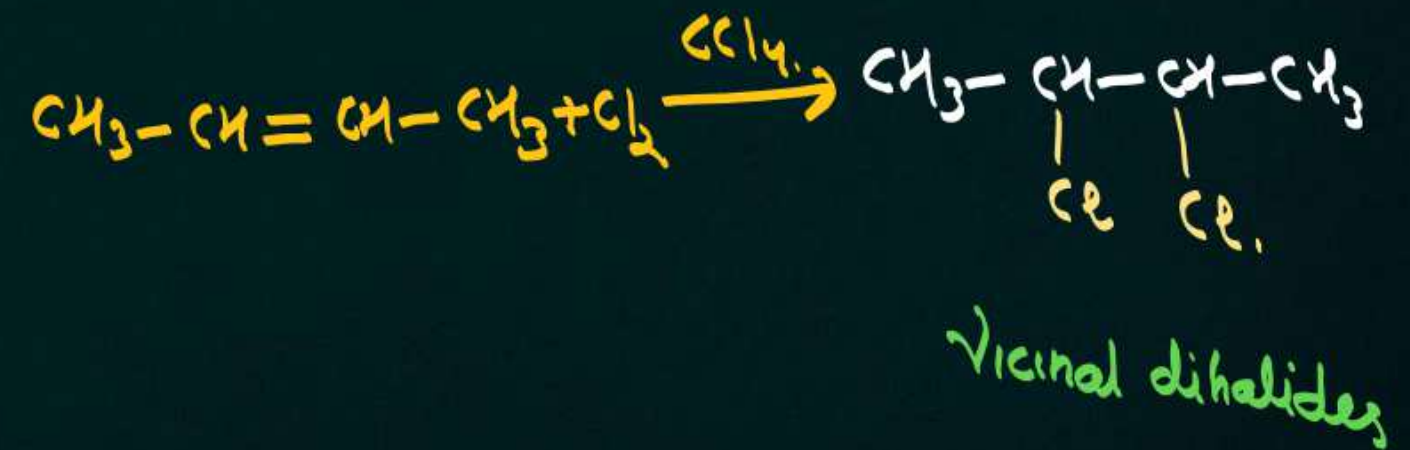
(a) Addition of Dihydrogen ^(H₂)

- Alkenes when treated with H₂ in the presence of finely divided Ni, Pd or Pt, give alkane as the product.

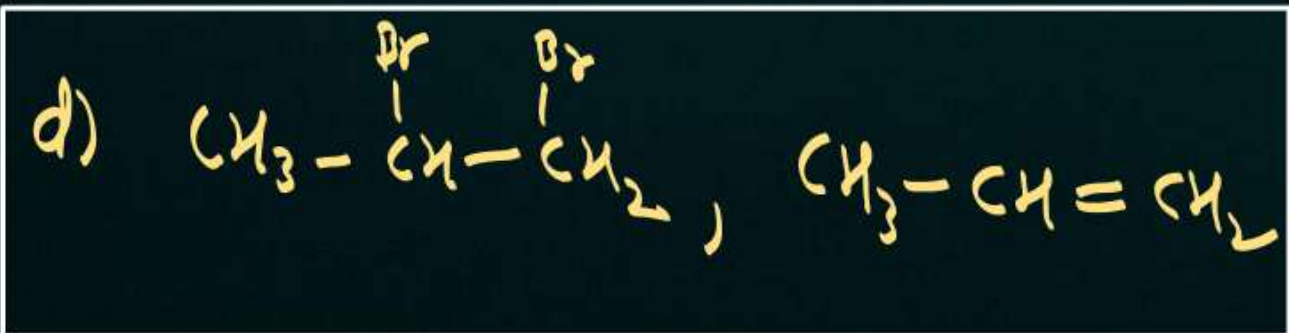
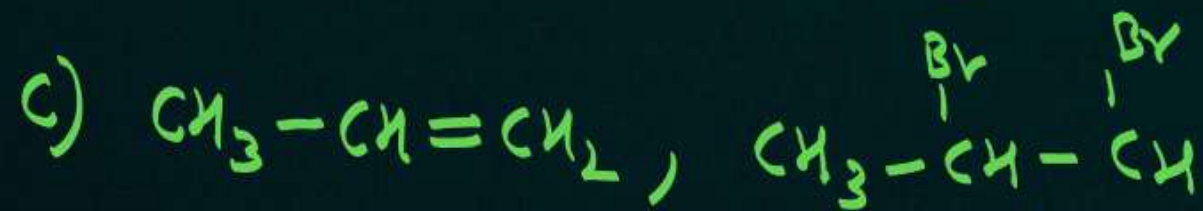
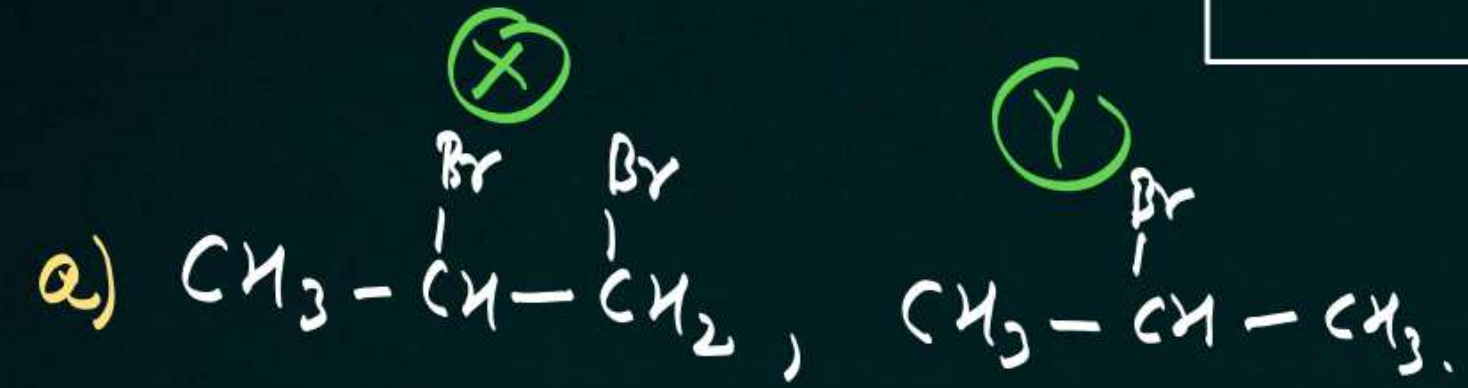
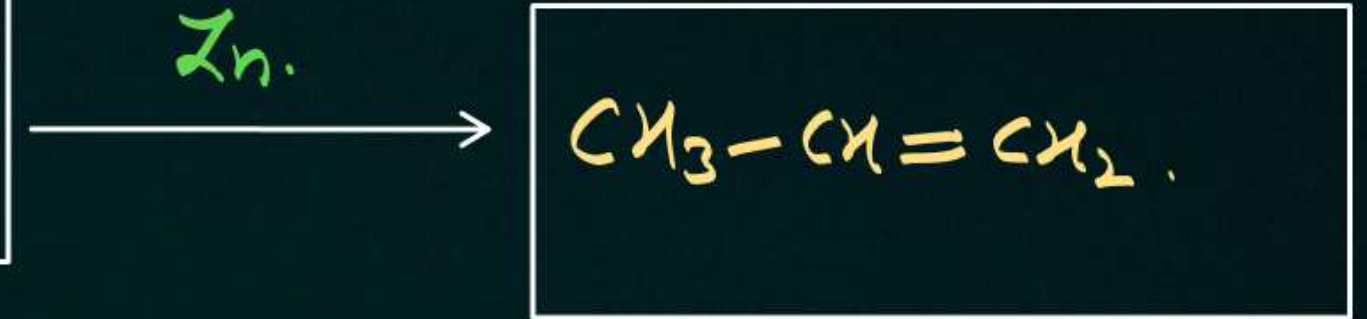
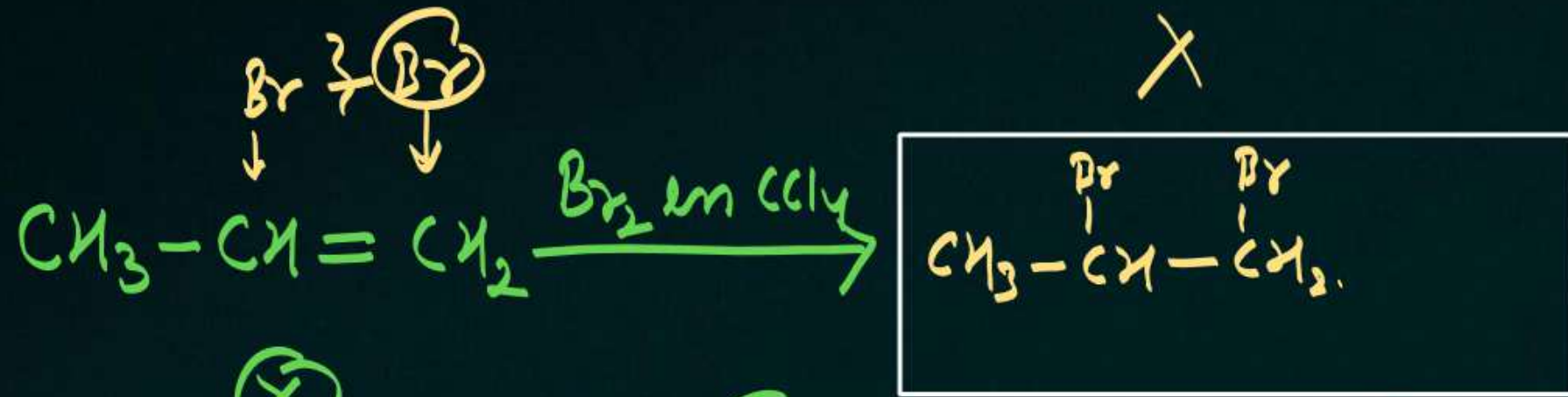


(b) Addition of Halogen ^(X₂)

- Alkenes give addition product, i.e. vicinal dihalides, when treated with halogens like Cl₂ or Br₂.



Q.



Chemical Properties of Alkenes

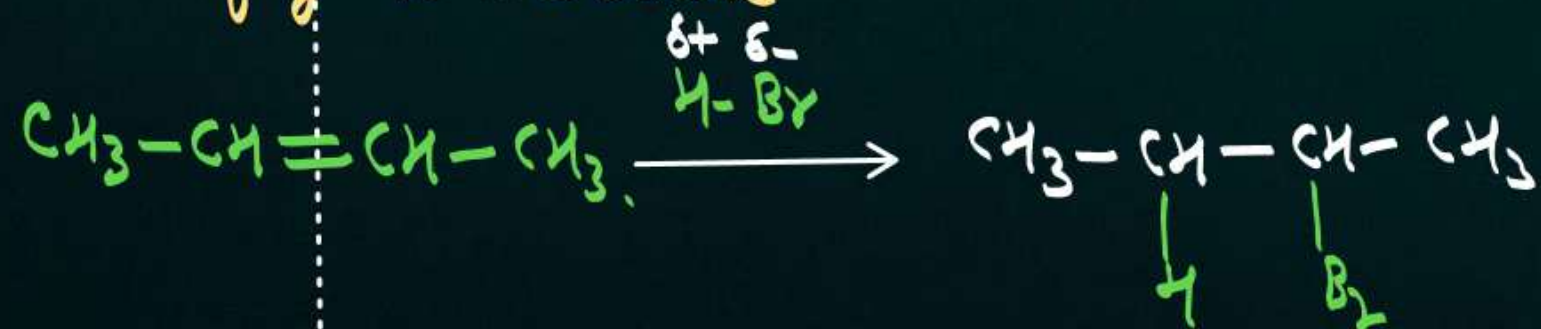


(c) Addition of Hydrogen Halides (HX)

- When alkenes react with hydrogen halides (HCl, HBr, HI), alkyl halides are obtained.
- The order of reactivity of hydrogen halides towards this reaction is

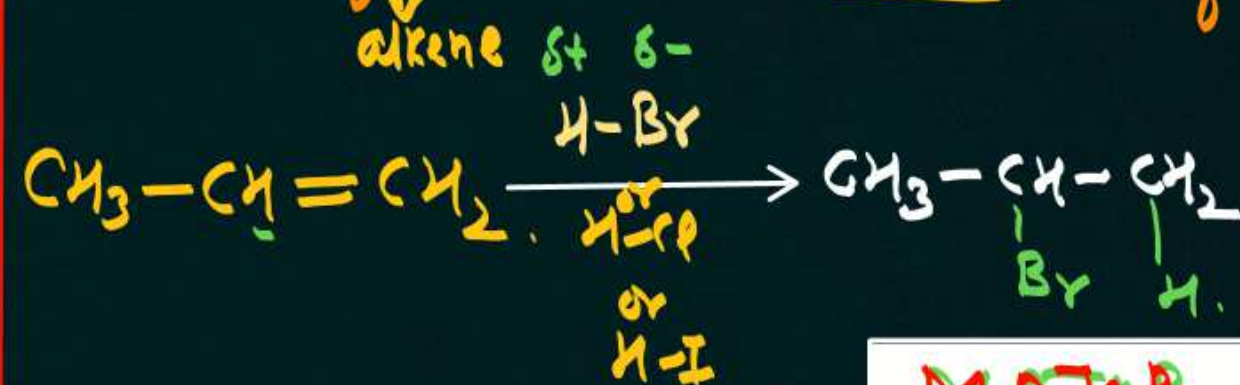


In case of symmetrical alkene



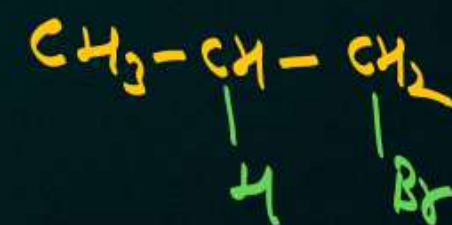
But in case of asymmetrical alkene, product is formed according to "MARKOVNIKOV RULE".

"It states that in case of Electrophilic addition reaction of alkene, negative part of reagent gets attached to that carbon of which has less no. of hydrogen"



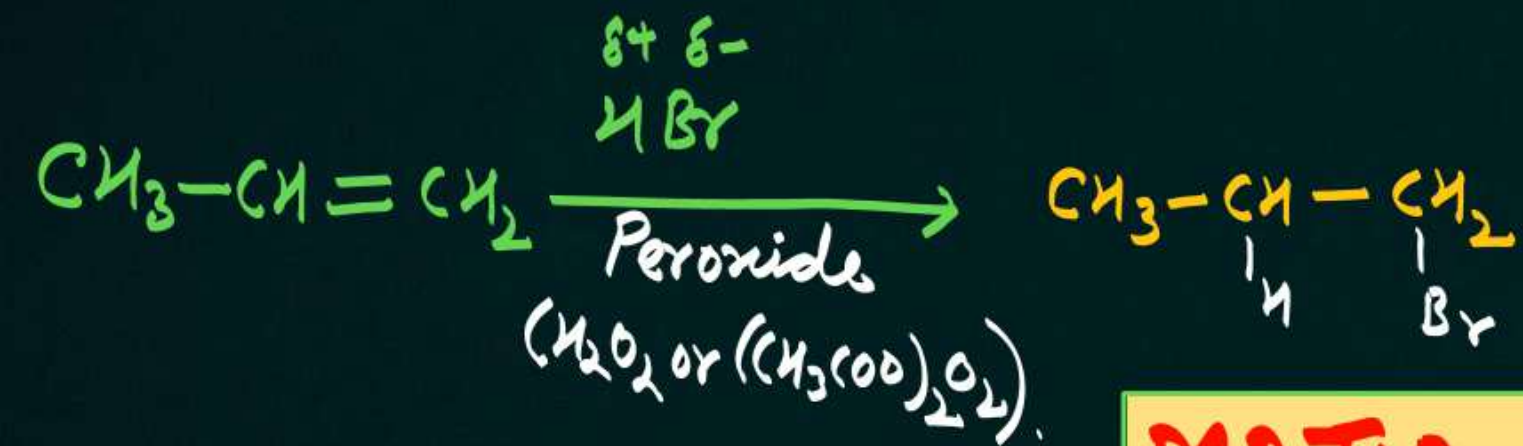
MAJOR

+



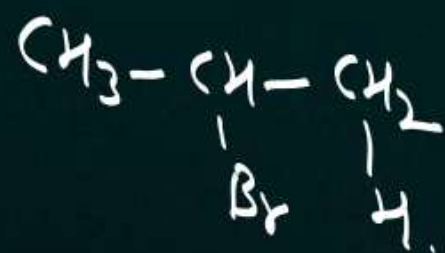
MINOR

But in case of addition of HBr to alkene, if reaction takes place in presence of peroxide, the product is formed according to "Antimarkovnikov Rule".



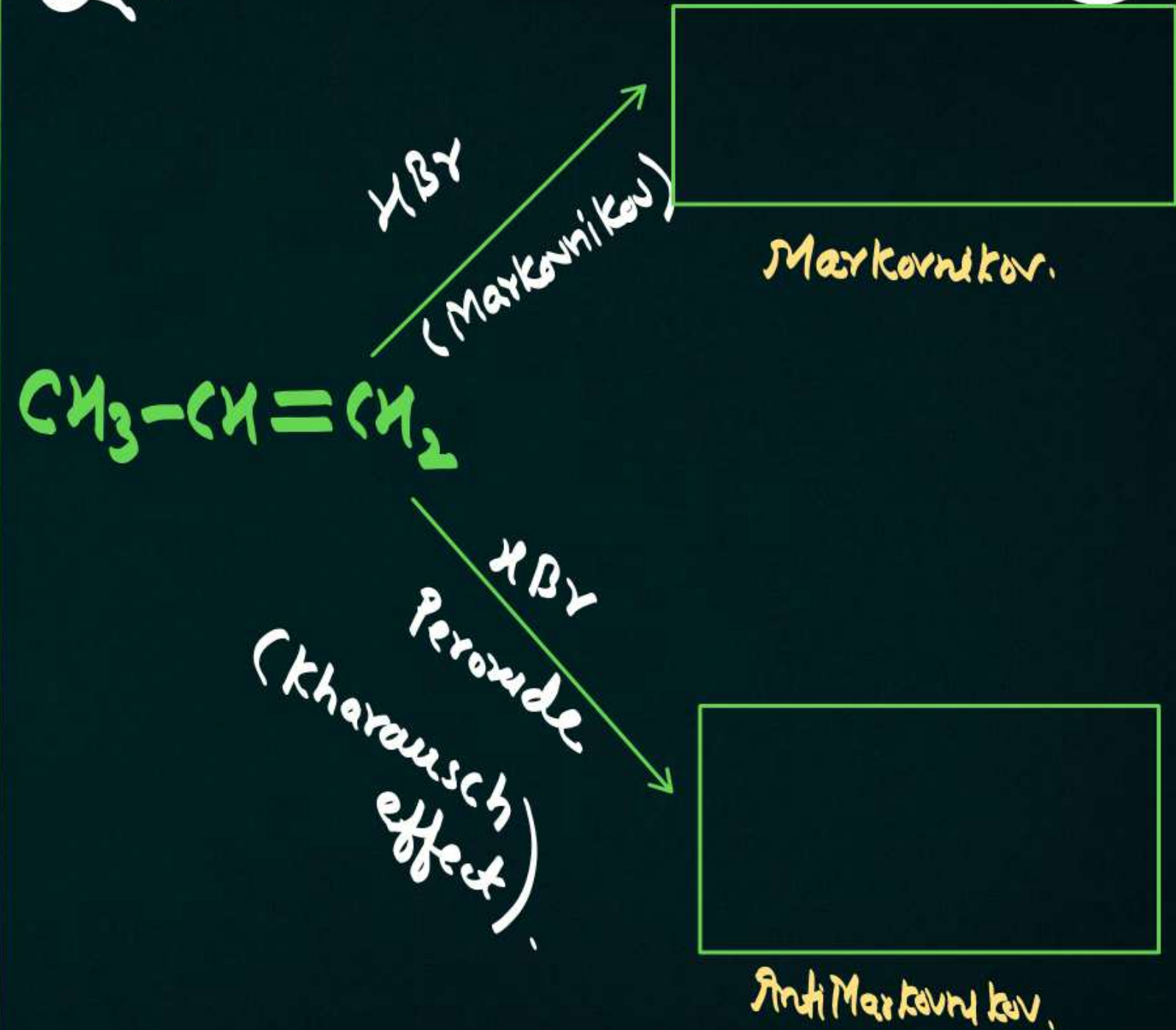
MAJOR

This effect is called as Kharasch effect or Peroxide effect.



MINOR

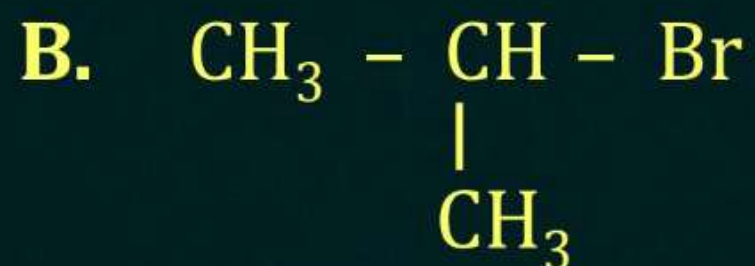
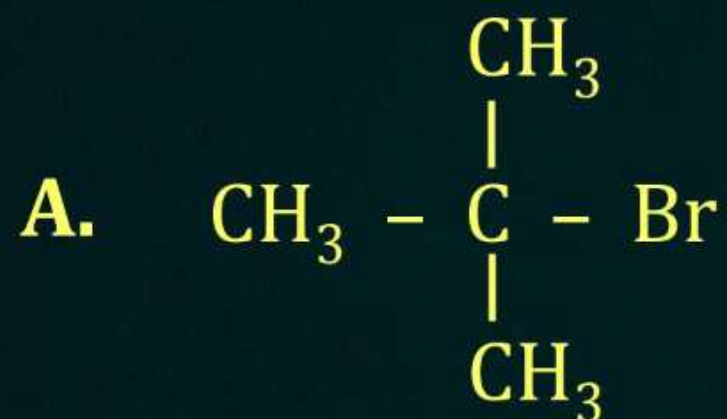
Q.



QUESTION



Arrange the following alkyl halides in decreasing order of the rate of β -elimination reaction with alcoholic KOH.



1 $A > B > C$

2 $C > B > A$

3 $B > C > A$

4 $A > C > B$

QUESTION



Which of the following has higher dipole moment; cis-2-butene and trans-2-butene?

1

cis-but-2-ene

2

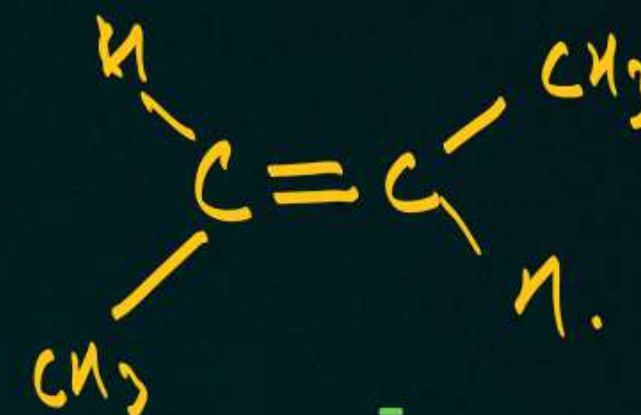
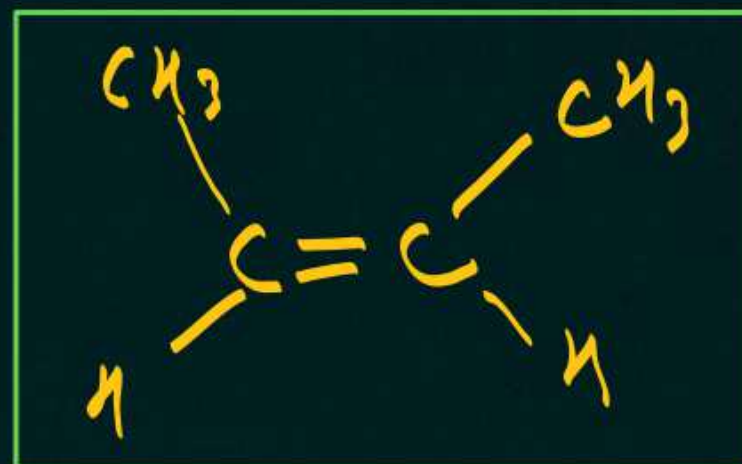
trans-but-2-ene

3

Both have same dipole moment

4

None of the above



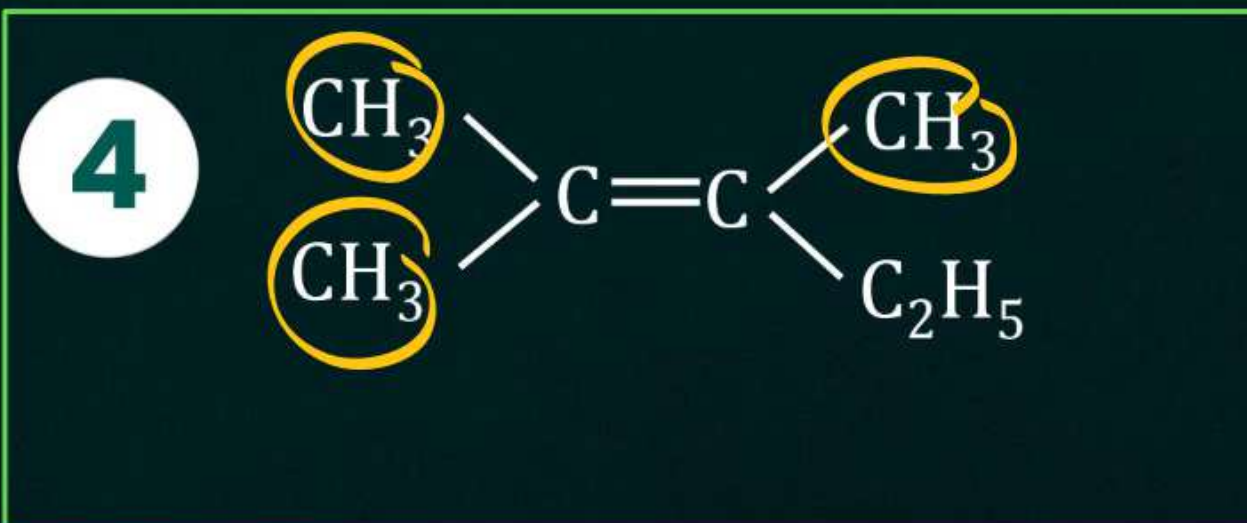
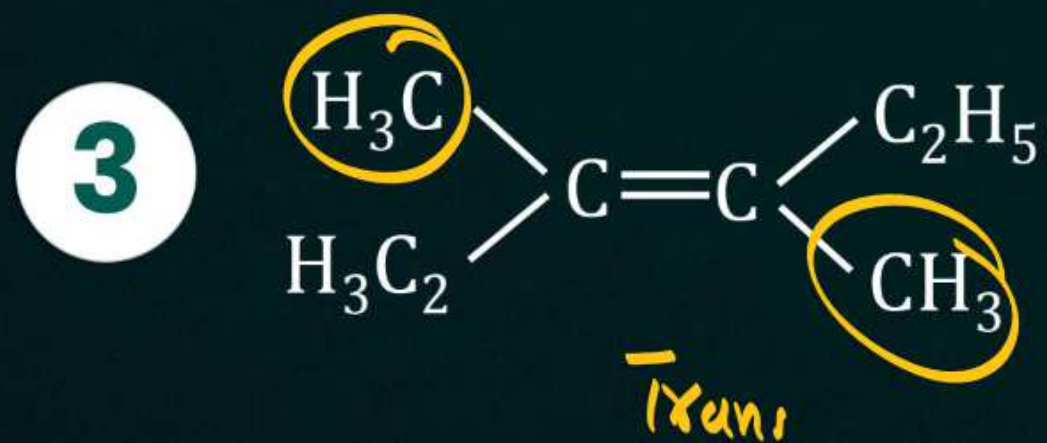
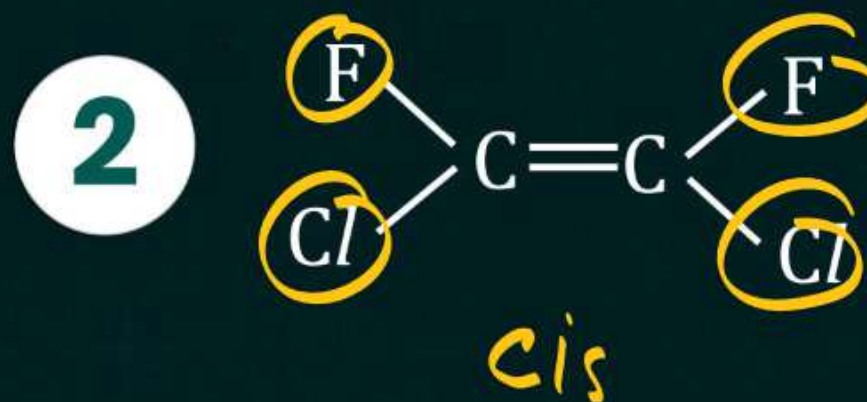
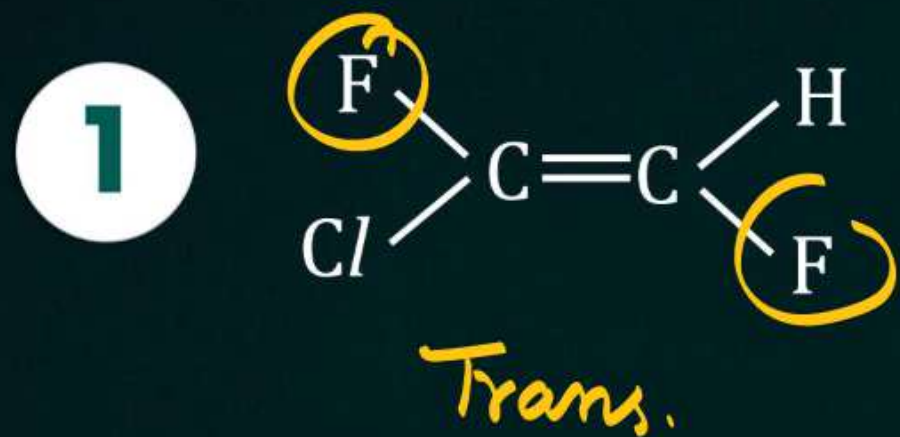
QUESTION

The major product formed when 3,3-dimethyl butan-2-ol is heated with concentrated sulphuric acid is

- 1 2,3-dimethylbut-1-ene
- 2 2,3-dimethylbut-2-ene
- 3 3,3-dimethylbut-1-ene
- 4 *cis* and *trans*-isomers of 2,3-dimethylbut-1-ene

QUESTION

Which of the following will not show **geometrical isomerism**?



1) Para Commando
Tanzeem.

2) Dr. Aleem.

3) Soliheen Lone.

Thank *You* #XUV
❤️